B.2: ten hosts, POWER / RESET only. Original case LEDs remain on motherboard.
5 V input -> 3.3 V logic; ESP32-S3FN8 + W5500 -> 74HC595 -> twenty isolated dry contacts
MC14541B counted timeout + independent ARM gate; no LED sensing or isolated DC/DC
S.3 compact layout; routing authorized after placement checks; no fabrication release

5 V input, protection and 3.3 V buck

檔案： power.kicad_sch

ESP32-S3FN8 / USB programming (1/2)

檔案： processor_1.kicad_sch

ESP32-S3FN8 / USB programming (2/2)

檔案： processor_2.kicad_sch

W5500 / HR911105A Ethernet (1/3)

檔案： ethernet_1.kicad_sch

W5500 / HR911105A Ethernet (2/3)

檔案： ethernet_2.kicad_sch

W5500 / HR911105A Ethernet (3/3)

檔案： ethernet_3.kicad_sch

Output latches / counted timeout (1/2)

檔案： outputs_1.kicad_sch

Output latches / counted timeout (2/2)

檔案： outputs_2.kicad_sch

Host 01: POWER / RESET only

檔案： host01.kicad_sch

Host 02: POWER / RESET only

檔案： host02.kicad_sch

Host 03: POWER / RESET only

檔案： host03.kicad_sch

Host 04: POWER / RESET only

檔案： host04.kicad_sch

Host 05: POWER / RESET only

檔案： host05.kicad_sch

Host 06: POWER / RESET only

檔案： host06.kicad_sch

Host 07: POWER / RESET only

檔案： host07.kicad_sch

Host 08: POWER / RESET only

檔案： host08.kicad_sch

Host 09: POWER / RESET only

檔案： host09.kicad_sch

Host 10: POWER / RESET only

檔案： host10.kicad_sch

Controller power and output-enable indicators

檔案： indicators.kicad_sch

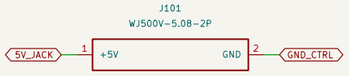
B.2 circuit — prototype review, not a manufacturing release

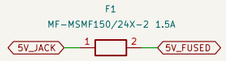
Sheet: /
File: rack-atx-controller.kicad_sch

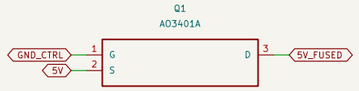
Title: 10-host rack ATX controller

Size: A3	Date:	**Rev: B.2**
KiCad E.D.A. 10.0.0		Id: 1/20

Regulated 5 V +/−5%, 2 A, terminal 1 positive. USB VBUS is NOT a power input.
Check PCB thermal/current loops and external PSU before energizing.







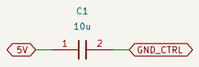


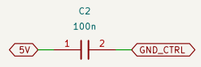




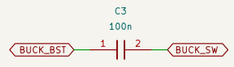


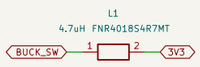


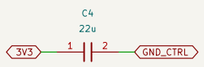














SCHEMATIC REVIEW — not a fabrication release

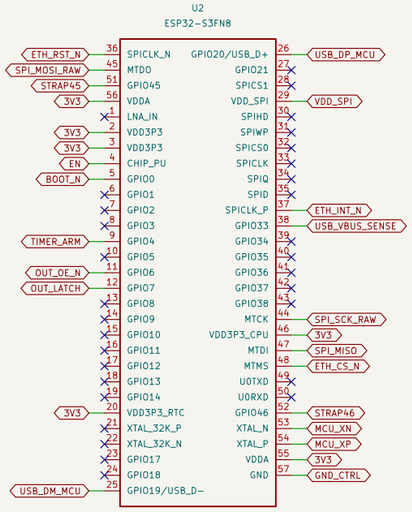
Rack ATX Controller / ddasdkimo

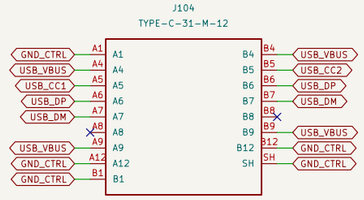
Sheet: /5 V input, protection and 3.3 V buck/
File: power.kicad_sch

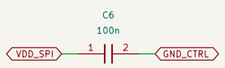
Title: 5 V input, protection and 3.3 V buck

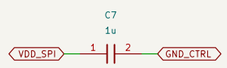
Size: A3	Date:	**Rev: B.2**
KiCad E.D.A. 10.0.0		Id: 2/20

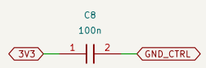
Wired Ethernet only. Leave LNA_IN open and never initialize Wi-Fi/BLE.
GPIO26..32 used by internal quad flash: do not route. GPIO45 pulled low selects 3.3 V flash.

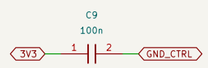


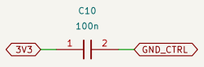


















SCHEMATIC REVIEW — not a fabrication release

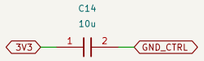
Rack ATX Controller / ddasdkimo

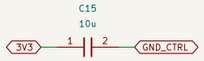
Sheet: /ESP32-S3FN8 / USB programming (1/2)/
File: processor_1.kicad_sch

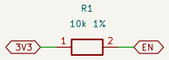
Title: ESP32-S3FN8 / USB programming (1/2)

Size: A3	Date:	**Rev: B.2**
KiCad E.D.A. 10.0.0		Id: 3/20

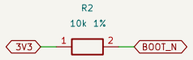
Wired Ethernet only. Leave LNA_IN open and never initialize Wi-Fi/BLE.
GPIO26..32 used by internal quad flash: do not route. GPIO45 pulled low selects 3.3 V flash.



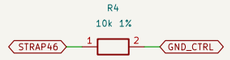








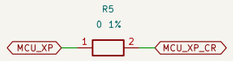






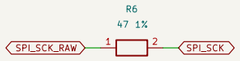


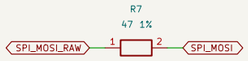


























SCHEMATIC REVIEW — not a fabrication release

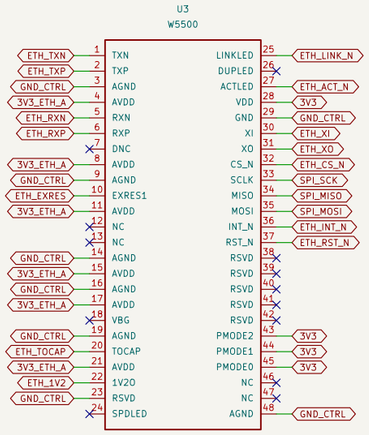
Rack ATX Controller / ddasdkimo

Sheet: /ESP32-S3FN8 / USB programming (2/2)/
File: processor_2.kicad_sch

Title: ESP32-S3FN8 / USB programming (2/2)

Size: A3	Date:	**Rev: B.2**
KiCad E.D.A. 10.0.0		Id: 4/20

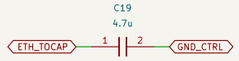
WIZnet RJ45 reference topology. HR911105A pin numbering differs from J1B1211CCD.
100-ohm differential MDI routing; crystal load caps need final frequency measurement.

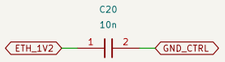


















SCHEMATIC REVIEW — not a fabrication release

Rack ATX Controller / ddasdkimo

Sheet: /W5500 / HR911105A Ethernet (1/3)/
File: ethernet_1.kicad_sch

Title: W5500 / HR911105A Ethernet (1/3)

Size: A3	Date:	**Rev: B.2**
KiCad E.D.A. 10.0.0		Id: 5/20

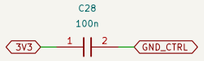
WIZnet RJ45 reference topology. HR911105A pin numbering differs from J1B1211CCD.
100-ohm differential MDI routing; crystal load caps need final frequency measurement.















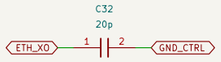
















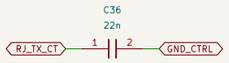




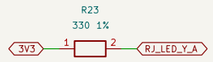


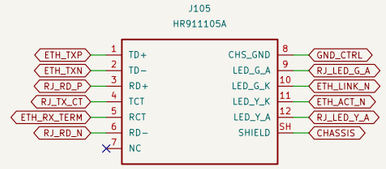












SCHEMATIC REVIEW — not a fabrication release

Rack ATX Controller / ddasdkimo

Sheet: /W5500 / HR911105A Ethernet (2/3)/
File: ethernet_2.kicad_sch

Title: W5500 / HR911105A Ethernet (2/3)

Size: A3	Date:	**Rev: B.2**
KiCad E.D.A. 10.0.0		Id: 6/20

WIZnet RJ45 reference topology. HR911105A pin numbering differs from J1B1211CCD.
100-ohm differential MDI routing; crystal load caps need final frequency measurement.



SCHEMATIC REVIEW — not a fabrication release

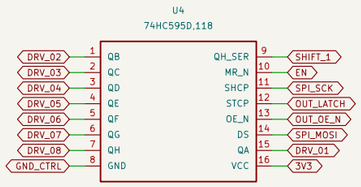
Rack ATX Controller / ddasdkimo

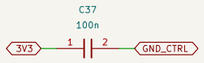
Sheet: /W5500 / HR911105A Ethernet (3/3)/
File: ethernet_3.kicad_sch

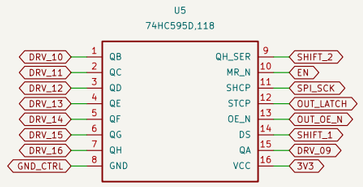
Title: W5500 / HR911105A Ethernet (3/3)

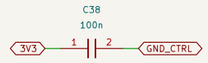
Size: A3	Date:	**Rev: B.2**
KiCad E.D.A. 10.0.0		Id: 7/20

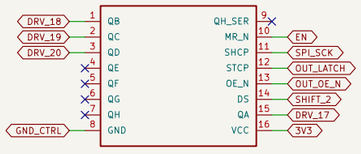
Default: ARM low, OE high, latches cleared before enabling. GPIO4/6/7 avoid boot UART pulls and GPIO1/2 glitches.
MC14541B: AR=1 disables automatic reset; MR pulled high until ARM. MODE=0, QSEL=1, A=B=1.
200k / 1nF C0G oscillator ~2174 Hz; 32768-count timeout ~15.07s. Release ARM immediately ends output.
ARM must be low >=10ms between commands. Hardware output requires both ARM and timer Q. No LED sensing.















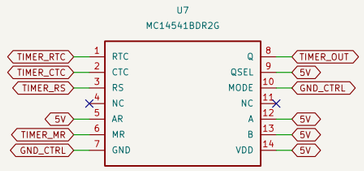
















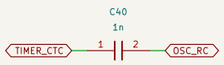


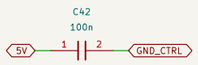
















SCHEMATIC REVIEW — not a fabrication release
Rack ATX Controller / ddasdkimo
Sheet: /Output latches / counted timeout (1/2)/
File: outputs_1.kicad_sch
Title: Output latches / counted timeout (1/2)
Size: A3 | Date: | **Rev: B.2**
KiCad E.D.A. 10.0.0 | Id: 8/20

Default: ARM low, OE high, latches cleared before enabling. GPIO4/6/7 avoid boot UART pulls and GPIO1/2 glitches.
MC14541B: AR=1 disables automatic reset; MR pulled high until ARM. MODE=0, QSEL=1, A=B=1.
200k / 1nF C0G oscillator ~2174 Hz; 32768-count timeout ~15.07s. Release ARM immediately ends output.
ARM must be low >=10ms between commands. Hardware output requires both ARM and timer Q. No LED sensing.

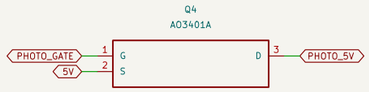














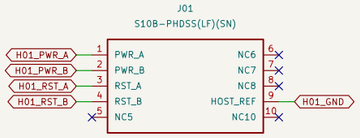


SCHEMATIC REVIEW — not a fabrication release
Rack ATX Controller / ddasdkimo
Sheet: /Output latches / counted timeout (2/2)/
File: outputs_2.kicad_sch

Title: Output latches / counted timeout (2/2)

Size: A3	Date:	**Rev: B.2**
KiCad E.D.A. 10.0.0		Id: 9/20

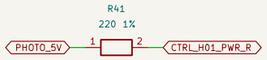
Two isolated normally-open contacts parallel the original case buttons.
No host LED sensing or isolated supply. Pin 9 remains this host reference for button ESD protection ONLY.
Do not join host references to controller ground or to one another. Pins 5..8 and 10 are NC.
Use verified motherboard signal names, not a universal F_PANEL physical pin map.





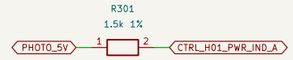


































SCHEMATIC REVIEW — not a fabrication release

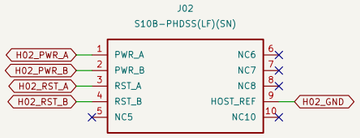
Rack ATX Controller / ddasdkimo

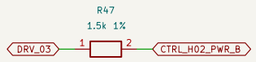
Sheet: /Host 01: POWER / RESET only/
File: host01.kicad_sch

Title: Host 01: POWER / RESET only

Size: A3	Date:	**Rev: B.2**
KiCad E.D.A. 10.0.0		Id: 10/20

Two isolated normally-open contacts parallel the original case buttons.
No host LED sensing or isolated supply. Pin 9 remains this host reference for button ESD protection ONLY.
Do not join host references to controller ground or to one another. Pins 5..8 and 10 are NC.
Use verified motherboard signal names, not a universal F_PANEL physical pin map.











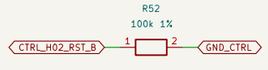






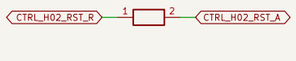


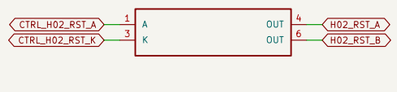








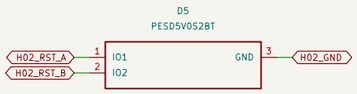






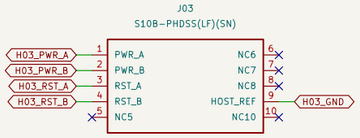






SCHEMATIC REVIEW — not a fabrication release
Rack ATX Controller / ddasdkimo
Sheet: /Host 02: POWER / RESET only/
File: host02.kicad_sch
Title: Host 02: POWER / RESET only
Size: A3 | Date: | **Rev: B.2**
KiCad E.D.A. 10.0.0 | Id: 11/20

Two isolated normally-open contacts parallel the original case buttons.
No host LED sensing or isolated supply. Pin 9 remains this host reference for button ESD protection ONLY.
Do not join host references to controller ground or to one another. Pins 5..8 and 10 are NC.
Use verified motherboard signal names, not a universal F_PANEL physical pin map.



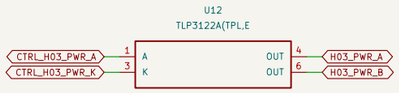












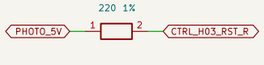


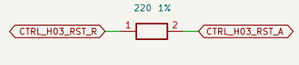
















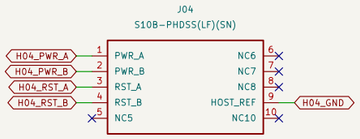


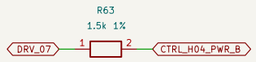




SCHEMATIC REVIEW — not a fabrication release
Rack ATX Controller / ddasdkimo
Sheet: /Host 03: POWER / RESET only/
File: host03.kicad_sch
Title: Host 03: POWER / RESET only
Size: A3 | Date: | **Rev: B.2**
KiCad E.D.A. 10.0.0 | Id: 12/20

Two isolated normally-open contacts parallel the original case buttons.
No host LED sensing or isolated supply. Pin 9 remains this host reference for button ESD protection ONLY.
Do not join host references to controller ground or to one another. Pins 5..8 and 10 are NC.
Use verified motherboard signal names, not a universal F_PANEL physical pin map.













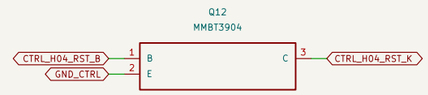






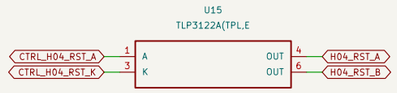














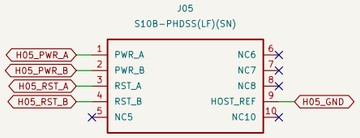


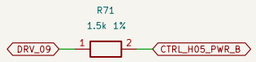




SCHEMATIC REVIEW — not a fabrication release
Rack ATX Controller / ddasdkimo
Sheet: /Host 04: POWER / RESET only/
File: host04.kicad_sch
Title: Host 04: POWER / RESET only
Size: A3 | Date: | **Rev: B.2**
KiCad E.D.A. 10.0.0 | Id: 13/20

Two isolated normally-open contacts parallel the original case buttons.
No host LED sensing or isolated supply. Pin 9 remains this host reference for button ESD protection ONLY.
Do not join host references to controller ground or to one another. Pins 5..8 and 10 are NC.
Use verified motherboard signal names, not a universal F_PANEL physical pin map.











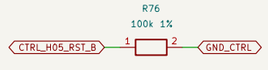






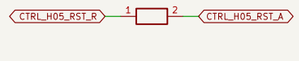
















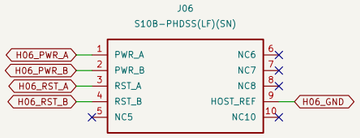






SCHEMATIC REVIEW — not a fabrication release
Rack ATX Controller / ddasdkimo
Sheet: /Host 05: POWER / RESET only/
File: host05.kicad_sch
Title: Host 05: POWER / RESET only
Size: A3 | Date: | **Rev: B.2**
KiCad E.D.A. 10.0.0 | Id: 14/20

Two isolated normally-open contacts parallel the original case buttons.
No host LED sensing or isolated supply. Pin 9 remains this host reference for button ESD protection ONLY.
Do not join host references to controller ground or to one another. Pins 5..8 and 10 are NC.
Use verified motherboard signal names, not a universal F_PANEL physical pin map.

















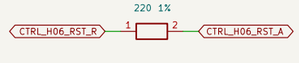


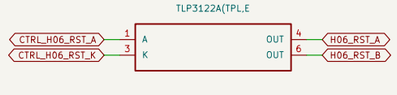


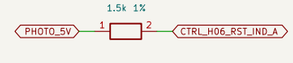












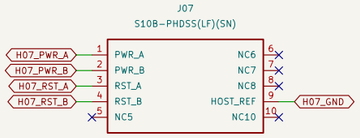






SCHEMATIC REVIEW — not a fabrication release
Rack ATX Controller / ddasdkimo
Sheet: /Host 06: POWER / RESET only/
File: host06.kicad_sch
Title: Host 06: POWER / RESET only
Size: A3 | Date: | **Rev: B.2**
KiCad E.D.A. 10.0.0 | Id: 15/20

Two isolated normally-open contacts parallel the original case buttons.
No host LED sensing or isolated supply. Pin 9 remains this host reference for button ESD protection ONLY.
Do not join host references to controller ground or to one another. Pins 5..8 and 10 are NC.
Use verified motherboard signal names, not a universal F_PANEL physical pin map.





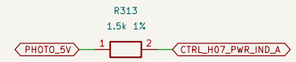






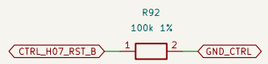








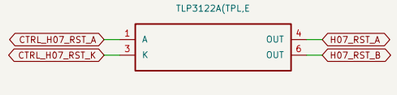


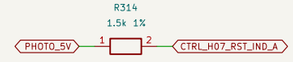






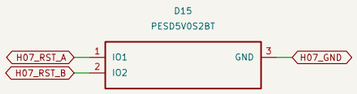












SCHEMATIC REVIEW — not a fabrication release

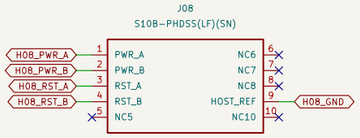
Rack ATX Controller / ddasdkimo

Sheet: /Host 07: POWER / RESET only/
File: host07.kicad_sch

Title: Host 07: POWER / RESET only

Size: A3	Date:	**Rev: B.2**
KiCad E.D.A. 10.0.0		Id: 16/20

Two isolated normally-open contacts parallel the original case buttons.
No host LED sensing or isolated supply. Pin 9 remains this host reference for button ESD protection ONLY.
Do not join host references to controller ground or to one another. Pins 5..8 and 10 are NC.
Use verified motherboard signal names, not a universal F_PANEL physical pin map.



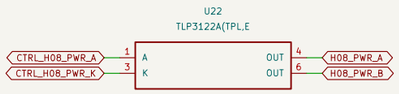








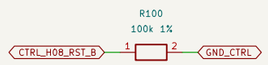


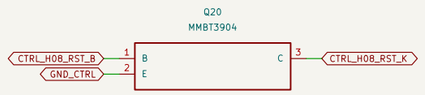






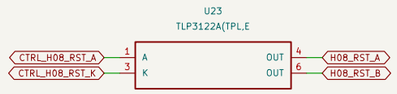














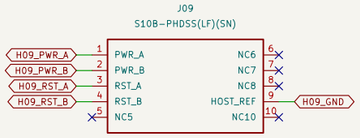


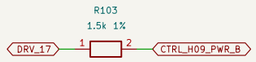




SCHEMATIC REVIEW — not a fabrication release
Rack ATX Controller / ddasdkimo
Sheet: /Host 08: POWER / RESET only/
File: host08.kicad_sch
Title: Host 08: POWER / RESET only
Size: A3 | Date: | **Rev: B.2**
KiCad E.D.A. 10.0.0 | Id: 17/20

Two isolated normally-open contacts parallel the original case buttons.
No host LED sensing or isolated supply. Pin 9 remains this host reference for button ESD protection ONLY.
Do not join host references to controller ground or to one another. Pins 5..8 and 10 are NC.
Use verified motherboard signal names, not a universal F_PANEL physical pin map.



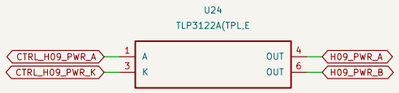








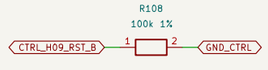










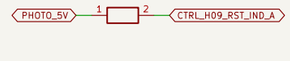












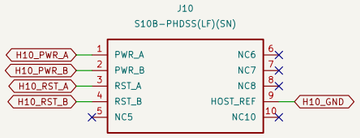


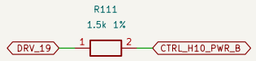


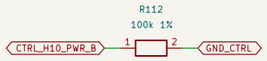


SCHEMATIC REVIEW — not a fabrication release
Rack ATX Controller / ddasdkimo
Sheet: /Host 09: POWER / RESET only/
File: host09.kicad_sch
Title: Host 09: POWER / RESET only
Size: A3 | Date: | **Rev: B.2**
KiCad E.D.A. 10.0.0 | Id: 18/20

Two isolated normally-open contacts parallel the original case buttons.
No host LED sensing or isolated supply. Pin 9 remains this host reference for button ESD protection ONLY.
Do not join host references to controller ground or to one another. Pins 5..8 and 10 are NC.
Use verified motherboard signal names, not a universal F_PANEL physical pin map.



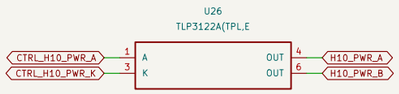






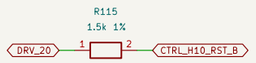












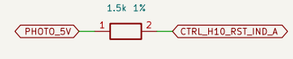












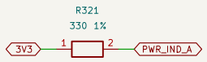






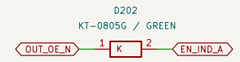
SCHEMATIC REVIEW — not a fabrication release
Rack ATX Controller / ddasdkimo
Sheet: /Host 10: POWER / RESET only/
File: host10.kicad_sch
Title: Host 10: POWER / RESET only
Size: A3 | Date: | **Rev: B.2**
KiCad E.D.A. 10.0.0 | Id: 19/20

PWR: 3V3 present. EN: OUT_OE_N low, output latches enabled.
Per-host red PWR/RST indicators follow the photorelay driver AND PHOTO_5V.
These indicate driver state, NOT actual contact closure or host power state.
RJ45 built-in LINK/ACT LEDs retained. All indicators use controller ground.









SCHEMATIC REVIEW — not a fabrication release
Rack ATX Controller / ddasdkimo
Sheet: /Controller power and output-enable indicators/
File: indicators.kicad_sch

Title: Controller power and output-enable indicators

Size: A3	Date:	**Rev: B.2**
KiCad E.D.A. 10.0.0		Id: 20/20